02-620 Week 10

Machine Learning for Scientists

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Dimensionality Reduction

Data in High Dimensional Space

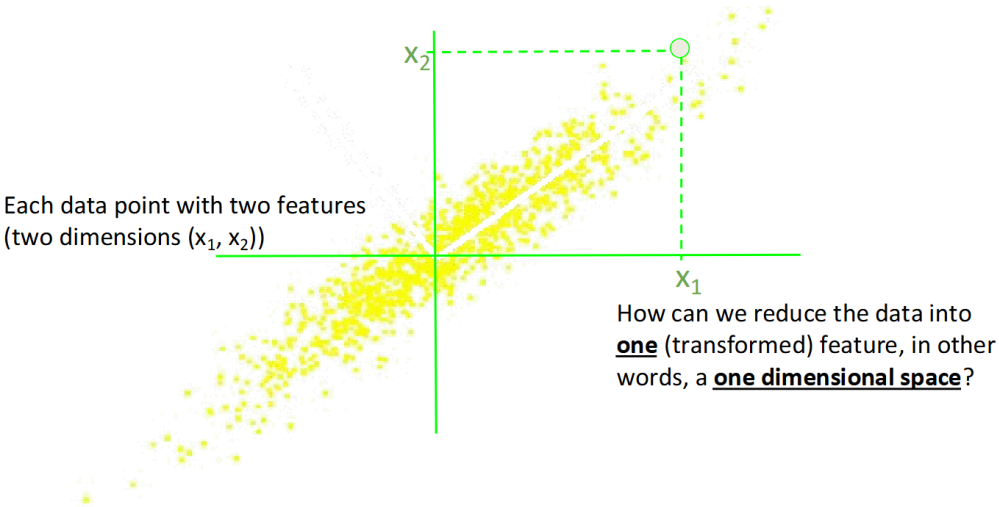
- High-dimensional data
  - Large number of features (large K)
  - Difficult to visualize
  - Difficult to fit a large model on data with many features
- Can we reduce the dimension of the data?
  - For easy visualization
  - For effective model learning

What is Dimensionality Reduction?

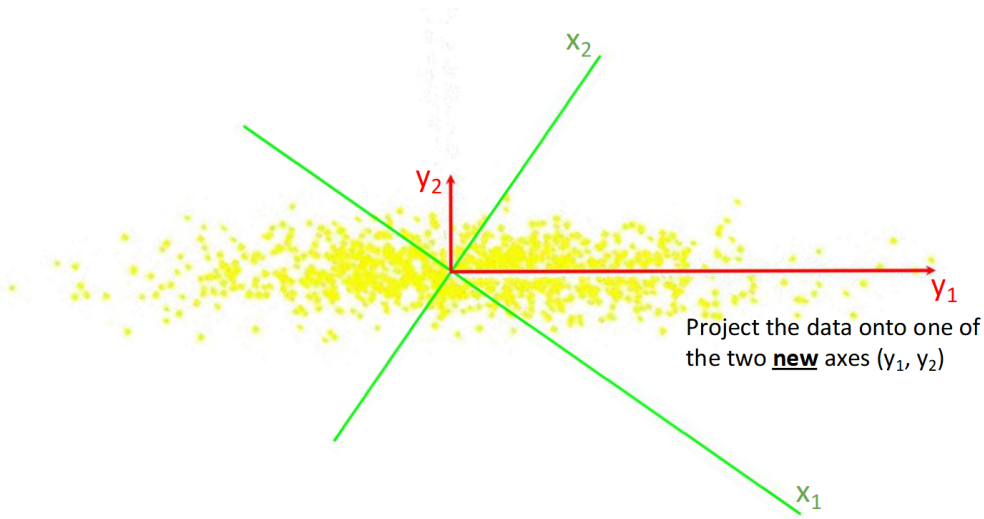
- Dimensionality reduction is a type of unsupervised learning
- We try to reduce the N sample $\times$ K feature training data to a N sample $\times$ M feature training data, where ($M << K$).
- Easy to visualize in 2 or 3 dimensional space
- Transformed features capture most important dimensions
- May capture the low-dimensional structure of the data

Principal Component Analysis (PCA)

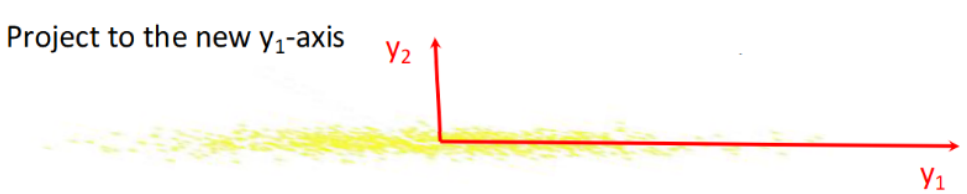
Consider reducing 2-dim data into 1-dim data.



Now, we can create new axes, y_1 and y_2 that also represent the data on the plane using some linear transformation. We rotate this new basis.



Finally, we project to one axes. In this case, projecting onto y_1 is better since the data has **higher variability** along y_1.



The result is that we now have less dimensions, but the data is still represented well (albeit not as well compared to by both features)

How to execute PCA

1. Find the principal components. (linear algebra: eigen-decomposition)
2. Project the data onto the principle components of your choice. (linear algebra: vector inner-product)

Aside: Eigen-decomposition

$$S = UDU^T$$

- S is a $K \times K$ symmetric square matrix
- U is a $K \times K$ square matrix
- D is a $K \times K$ diagonal matrix

In code, we can do this by calling the `numpy.linalg.eigh` function in Python.

- U and D are our eigenvalues and eigenvectors, respectively.
- $U = [u_1, \cdots, u_K] \in \mathbb{R}^{K \times K}$. $u_1, \cdots, u_K \in \mathbb{R}^K$
- $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$. $\lambda_1, \cdots, \lambda_K \in \mathbb{R}$

PCA Step 1: PCs via Eigen-Decomposition

- Given $N \times K$ matrix $X = \begin{bmatrix} X_1^T \\ \cdots \\ X_N^T \end{bmatrix}$, reduce dimension from K to M
- Mean center the data
  1. Compute $K \times K$ covariance matrix, $S = \frac{1}{N} X^T X$
  2. Eigen-decomposition to obtain eigenvalues and eigenvectors
     - $s = UDU^T$, where $U = [u_1, \cdots, u_k]$, $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$
     - Pick the M eigenvectors (principal components) with the largest eigenvalues

PCA Step 2: Project data onto PCs

$$x^T u_1 = \|x\| \|u_1\| \cos(\theta) = \|x\| \cos(\theta)$$

- u_1 is a unit vector of a principal component.
- Remember to set the center of the data to the origin!

PCA: Putting everything together

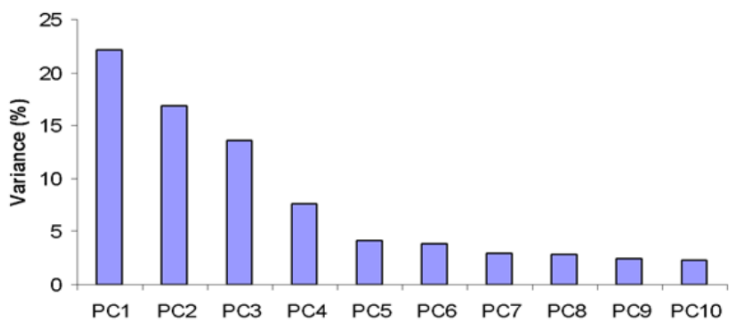
- Given $N \times K$ data matrix $K = \left[X_1^T \cdots, X_N^T\right]$, reduce dimension from K to M
- Mean center the data
- Step 1
  - Compute $K \times K$ covariance matrix $S = \frac{1}{N} X^T X$
  - Eigen-decomposition to obtain eigenvalues and eigenvectors
    - $S = UDU^T$, where $U = [u_1, \cdots, u_K]$, $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$
  - Pick M eigenvectors (principal components) with the **largest eigenvalues**
- Step 2
  - Project data onto the **top** M **eigenvectors** $X_{p,i}^T = [X_i^T u_1, \cdots, X_i^T u_M]$

How Many Principal Components (PCs) M?

- The k-th eigenvalue: variance of projected data onto the k-th PC.
- The **first PC** retains the greatest fraction of **variation** in the sample
- The k-th PC retains the k-th greatest fraction of variation in the sample

Therefore,

- For K original dimensions, there are K eigenvalues and vectors. So K PCs.
- **Ignore the compoennts of lesser significance** with lower variation in data
- Strategy: Plot the eigenvalues, use only the PCs with reasonably high eigenvalues.



PCA Summary

$K \times K$ covariance matrix $S = \frac{1}{N} X^T X$

- Eigen-decomposition of $S = UDU^T$, where $U = [u_1, \cdots, u_K]$, $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$, and λ_i's are sorted in descending order (λ_1 is the largest).
- $\sqrt{\lambda_k} u_k$ is the k-th PC, and u_k is called the k-th PC direction
  - Variance explained λ_k
- PC scores: projecting data onto the PC directions: $X^T_{\text{proj},i} = [X_i^T u_1, \cdots, X_i^T u_K]$
  - $\text{Var}(Xu_k) = \frac{1}{N} \sum_{i=1}^N (X_i^T u_k)^T = u_k^T (\frac{1}{N} X^T X) u_k = u_k^T S u_k = \lambda_k$
- Select top M PCs to explain a given percentage of total variance $\sum_{k=1}^K \lambda_k$

PCA as an Optimization

- Projection of vector x onto an axis u is $x^T u$
- X is a mean-centered $N \times K$ data matrix for N samples, K features
- Find a unit vector u such that the **variance of projected values is greatest**
  - Maximize $\frac{1}{N} u^T X^T X u$, such that $u^T u = 1$ (e.g, u is a unit vector).
  - Recall that

$$\text{Var}(X^T u) = \frac{1}{N} \sum_{i=1}^N (x_i^T u)^2 = \frac{1}{N} u^T X^T X u$$

How do we solve this?

- Construct Lagrangian
$$\frac{1}{N}u^T X^T X u + \lambda(1 - u^T u)$$
- Differentiate the Lagrangian with respect to u and set it to zero
$$\frac{1}{N}X^T X u - \lambda u = 0$$
- Or equivalently,
$$Su - \lambda u = 0$$
where $S = \frac{1}{n}X^T X$: covariance matrix
- As $u \neq 0$, then u must be an eigenvector of S with eigenvalue λ
- $u_1^T S u_1 = \lambda_1$ is the largest possible value. Thus, u_1 is the first PC direction. (Proof below)
- Given u_1, $u_2^T S u_2 = \lambda_2$ is the largest possible value across all directions orthogonal to u_1. Thus, u_2 is the second PC direction.
- Given $u_1, u_2, u_3^T S u_3 = \lambda_3$ is the largest possible value across all directions orthogonal to u_1, u_2, thus u_3 is the third PC direction...
- etc.

Proof that λ_1 is the largest possible value

- Consider any $u \neq u_1$ with $u^T u = 1$.
- Write $u = \sum_{k=1}^{K} a_k u_k$, then $u^T u = \sum_{k=1}^{K} a_k^2 = 1$.
- Furthermore, $u^T S u = \left(\sum_{k=1}^{K} a_k u_k\right)^T S \left(\sum_{k=1}^{K} a_k u_k\right) = \sum_{k=1}^{K} a_k^2 \lambda_k$
- This result is $< \lambda_1 \sum_{k=1}^{K} a_k^2 = lambda_1 = u_1^T S u_1$

Dimensionality Reduction of PCA

PCA leads to dimensionality reduction such that

- The variability in data are preserved in the lower dimensions
- For multiple correlated features (redundant features), remove the correlated dimensions
  - Focus on uncorrelated data
  - Note PCs are orthogonal!

Applying PCA to Images

Suppose you have 361×261 pixel images, and 83781 dimensional data. In other words, every image in the dataset can be represented perfectly as a vector of integers over the 83781 dimensions.

- In this case, your principal components are images!
- Flatten all the images into vectors and do the same process.
- Project the original image into 4 dimensions corresponding to the four PCs.

Summary

Principal Component Analysis (PCA)

- A type of unsupervised learning
- Principle: reduce the dimension of the original data such that the variability in the original data are preserved in the lower dimension
- In practice: use eigen decomposition
  - Top eigen values: the dimension that captures the most variability
  - Eign vectors with top eigen values: dimensions to be used to describe the lower dimensional data